



ANNUAL POLICY DIALOGUE 2026

CASE STUDY

**GOVERNANCE OF GLOBAL
DIGITAL PUBLIC GOODS
FOR DIGITAL PUBLIC
INFRASTRUCTURES**

Key Messages

- The crisis of global public goods governance and the trust deficit afflicting multilateralism are not the result of the current rupture of the international order, which cannot be resolved by restoring its institutions. Rather, the geopolitical upheaval exposes the uneven distribution of the benefits of the hegemonic order while it held.
- Digitalisation and datafication are deeply implicated in intensifying globalisation and the current geopolitical crisis. Global governance failures—both to realise the internet as a foundational digital public good at the national level and to prevent the private enclosure of data, the central resource shaping value creation and power—have driven rising inequality and enabled the accumulation of wealth and political influence now fuelling the great-power struggle for AI dominance.
- Data’s non-rivalrous nature makes it the foundational public good of digital public infrastructure. Openness is an instrumental governance mechanism to prevent private enclosure – not a primary public good in itself – and holds only when backed by enforceable frameworks that prevent excludability from being reimposed through proprietary control or market concentration.
- Trust is not a technical or legal feature of digital systems – it is the outcome of political choices that build institutional legitimacy. Trust derives from the ‘publicness’ of the institutions that govern it, not from technical standards in and of themselves.
- Financial sustainability is the central governance question for global public goods and arguably the greatest failure in the case of digital public goods. It is directly connected to sovereign agency and legitimacy.
- Under the right structural conditions, DPI could address longstanding developmental challenges of state formation, public value creation, and social good outcomes. Deployed as a techno-solution without sovereign agency, financial sustainability, or political legitimacy, it reproduces the enclosures and dependencies it was designed to overcome.
- Regional DPI – encompassing shared policy frameworks, infrastructure, human capital, and single-market integration – may offer the most viable pathway for the Global South, provided that institutional endowments are sufficiently developed to govern dynamic data systems and underlying public infrastructures.

Note

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1. Global Context: The Structural Entanglement of Geopolitical Crisis and Advanced Data-Driven Technology

Club de Madrid's 2026 Policy Dialogue proceeds from a foundational insight: that restoring democratic accountability and effectively providing global public goods are two sides of the same coin, linked by a double-sided causality in which trust enables effective provision and effective provision builds trust. This paper takes that insight as its departure point, arguing that the multilateral trust deficit predates the current rupture: it was built into the foundations.

Presenting the hegemonic order's accomplishments – financial, security, and trade stability – implies that its benefits accrued equitably, and that the recent loss of trust in its institutions requires restoration. Neither is true. The failure to realise global public goods across the Global South was not incidental to the international order but an outcome of the structural inequalities it simultaneously created and reproduced.

For much of the world, the multilateral system was a framework nations had no meaningful role in constructing but from which there was no viable exit. The underdevelopment of formally sovereign states, shaped by postcolonial economic arrangements and elite political settlements, constitutes a permanent condition of that order rather than a deviation from it (Acemoglu and Robinson, 2013; Khan, 2018). The Non-Aligned Movement, the G77, and more recently BRICS+ represent successive plurilateral efforts to navigate, rather than endorse, that reality. The shift from rules-based to openly power-based international relations therefore registers differently depending on structural position: for states that benefited most from rule enforcement, it is a genuine rupture; for states never reliably protected by those rules, it is a change in the character of their exposure, not its existence. The rise of more equitable and inclusive plurilateral coalitions – organised around common interests and universalist values capable of being realised in practice – also opens opportunities for states and peoples marginalised under the unipolar order.

The governance of digital public goods cannot be understood outside this broader crisis in global public goods provision – but neither can it be treated as simply one instance of that crisis. Digitalisation and datafication are deeply implicated in both the intensification of globalisation and the conditions that require global governance, as well as in the erosion of the cooperative conditions necessary to achieve it (Kaul, Grunberg, and Stern, 1999). International institutions that formed to coordinate global digital governance increasingly struggle to manage transboundary data flows, digital infrastructures, and AI systems whose reach and consequences far exceed their jurisdictional mandates. The digital and data domain requires governance, but it is simultaneously a force actively reshaping the international order within which all global public goods – health, water, climate action – must be governed.

Despite two decades of multilateral effort, the global digital commons – spectrum, the internet, data – have never been equitably governed. Shaped from their inception by powerful states and, increasingly, private corporations able to exploit multilateral forums to their own ends, they have always been rationed along lines of power and wealth.

Multilateral institutions have not only failed to govern digital public goods adequately but, in key instances, have actively worsened developmental outcomes. The WTO moratorium on digital trade taxes has transferred value from public to private hands at scale (Banga, 2022). WIPO has progressively tightened intellectual property rules through TRIPS, enclosing the knowledge commons (Frischmann et al., 2014; Hess and Ostrom, 2007; Stiglitz, 1999), while those same rules have proven incapable of preventing AI corporations from appropriating copyrighted and communally produced knowledge as training data (Chen, 2024). Sovereign indebtedness and access to concessional finance from the IMF and World Bank have consistently been conditioned in ways that further constrained the agency of states seeking to build their own digital infrastructure.

Big Power Struggle for AI Dominance

Big power competition is no longer primarily organised around oil or trade advantage but around control of the dominant means of production in advanced capitalism – the full value chain from rare earth minerals and data extraction to hypercompute power and specialist skills. Whoever controls the chokepoints at each layer exercises determining influence over the economic and governance capacities of the states that depend on them. Scientific and technological domains – AI, semiconductors, data infrastructure, technical standards – have become primary arenas of interstate contestation, with economic instruments – industrial policy, export controls, sanctions, aid conditionality – deployed as tools of strategic advantage (GESDA, n.d.; Palan, 2013).

The unchecked dynamic of creative destruction has not only displaced previous technologies and regulatory frameworks but has circumvented the sovereign member-state international system itself (Birhane et al., 2026; Gillwald, 2026). The international discourse of the 2010s – centred on the free flow of data and resistance to digital sovereignty claims – has given way to a strategic logic in which data infrastructure and artificial intelligence are classified as critical assets in national security strategies and industrial policy (Gillwald, 2026; Rohozinski, 2026).

The power asymmetries that shaped whose interests the multilateral system served are the same asymmetries that now determine who captures the benefits of digital infrastructure and who bears its costs. Benefits flow to a small number of corporations in a small number of countries; costs – of exclusion, data extraction, and the erosion of governance capacity – are borne by the populations whose data, labour, and resources generate those benefits. The consequential decisions shaping global markets, disrupting labour, and determining the fiscal capacity of governments are made by corporations accountable to neither the populations they affect nor the governments nominally responsible for their welfare.

The challenge for global public goods governance is therefore not simply technical or sectoral – it is structural. Recognising digital public infrastructure and the public goods embedded within it as a response to governance and market failures – failures produced by a decade of platformisation and now compounded by advanced data-driven systems built on unprecedented concentrations of ownership, wealth, and political influence – is the necessary starting point.

In the multilateral and donor-funded playbooks that dominate the DPI discourse and the technical solutions being rolled out across the global South, “...the transparency gaps limit public understanding and undermine democratic oversight, reducing the ability of citizens, civil society, journalists, and oversight bodies to scrutinise how these systems function and whose interests they serve. In many cases, DPI systems are developed and deployed with minimal proactive disclosure about who built them, how data flows between agencies, how decisions are made algorithmically, and how personal data is governed. Without such visibility, technical complexity becomes a shield that insulates public infrastructure from accountability, even when real-world harms, such as exclusion errors or privacy breaches, occur.” (Dube 2026)

The struggle over digital goods and infrastructure cannot be understood outside the broader geoeconomic competition now defining the global order, and it will not be resolved by frameworks that treat it as anything less.

This has fundamentally shifted the practice of sovereignty. Traditional notions premised on territorial control have been eroded by digital infrastructure whose reach transcends national jurisdictions. Uneven datafication works through spatial restructuring that simultaneously bypasses and reinforces territorial control, ranking populations as valuable or disposable, and enabling monopolistic platform control to extract surplus from peripheral regions (Akbari, 2026). Digital sovereignty is a daily practice requiring infrastructure ownership, regulatory capacity, and political will – and the conditions for that practice are being systematically dismantled (Razzano, 2026). Vendor lock-in, proprietary standards, and contractual dependencies strip states of the agency sovereignty requires. Dominant technology

firms have reframed sovereignty not as something states exercise over platforms but as something provisioned by them – ‘sovereignty-as-a-service,’ emptied of political content (Grohmann and Barbosa, 2024). For states in the Global South, sovereignty becomes something they can subscribe to but never own.

Reimagining a more just global order cannot mean restoring prior architecture. It requires institutional design that locates the multilateral system within the global political economy, including the inequalities it reproduces. Club de Madrid’s recognition of the two-way causality between trust and the effective provision of global public goods is the correct starting point – and must be the foundation of a new, equitable, and just international order. Sovereignty, legitimacy, and financial sustainability are inseparable from that project.

The UN multilateral system has been structurally constrained by design, and reforms in digital governance have been co-opted by powerful vested interests to prevent the creation of the ‘invisible trust infrastructure’ that genuine global commons governance would require. This has been achieved by stripping concepts of sovereignty and international cooperation of their political content, systematically decoupling democracy from innovation, and reducing governance to technical standards no longer set by multilateral bodies. The real challenge is building a new global trust infrastructure: grounded in accountability to the governed rather than deference to the powerful, and capable of earning trust rather than assuming it. That is harder than restoring institutions that reflected and reinforced structural inequality – but it is the only approach that explains why the old order failed as many as it served, and the only foundation on which the governance of digital public goods can credibly be built. This paper proposes that recoupling democracy with the governance of advanced data-driven technologies and reclaiming the building of publicly controlled digital infrastructures may counter the globally devastating effects of unchecked, unaccountable private enclosure of digital resources and data, which will otherwise remain critical fault lines in developing a new and just global order.

2. Analytical Framework: Public Goods Theory Applied to Digital Infrastructure

Revisiting public goods theory in relation to the global governance of digital public goods is a necessary corrective to the conflation of public goods with the related but distinct concepts of public interest, public value, and anything broadly perceived as ‘good for the public.’ Public goods theory identifies the structural logic of market failure in digital infrastructure and the challenges of private enclosure of its core contemporary instance – data. Samuelson’s (1954) twin criteria – non-excludability and non-rivalry – explain why markets systematically underprovide: non-excludability prevents cost recovery through sale, while non-rivalry means efficiency requires a zero price, extinguishing private investment incentives. Three refinements extend this analysis to the digital realm. Musgrave’s (1959) concept of merit goods supplies the normative rather than merely technical basis for public intervention – reflected in successive UN framings of broadband and internet access as goods requiring universal provisioning to realise human rights and democratic participation (UN Human Rights Council, 2011; 2016) – even where market failure is incomplete. Buchanan’s (1965) club goods framework – excludable but non-rival up to congestion – captures the intermediate position of most digital network infrastructure, including undersea cables. Olson’s (1965) formalisation of free-rider logic explains chronic underprovision across borders and generations, making voluntary contribution inadequate to resist enclosure – which is why, despite a powerful Open movement, some have argued that openness must in certain cases be mandatorily imposed on digital infrastructure to preserve its public good character (Gillwald, 2019).

Publicness as Governance Achievement

The touchstone for much of the subsequent research on public goods, pool resources, and commons draws on the work of Elinor Ostrom. Her central contention – that ‘publicness’ is not a natural property of a resource but an institutional outcome, the product of deliberate governance choices (Ostrom, 1990) – is foundational. In the digital realm, open-source software, scientific literature, and digital datasets are threatened not by overuse but by progressive privatisation through intellectual property extension and commodification – a new enclosure converting functionally public resources into proprietary goods at direct cost to social welfare (Hess and Ostrom, 2003; 2007). Internet governance, platform regulation, and data policy are therefore continuous institutional contests over the degree of publicness, not technical determinations of resource characteristics (Kaul et al., 1999; 2003). Publicness must be actively maintained against structural tendencies toward enclosure; it cannot be asserted once and assumed to hold (Best and Gheciu, 2014; Cammaerts and Mansell, 2020; Gillwald, 2021).

The Distinct Value Logic of Digital and Data Infrastructure

Digital infrastructure generates value through recombination rather than transaction. Where physical networks are constrained by geography and congestion, digital infrastructure increases in value with each additional participant, dataset, or application (Brynjolfsson and McAfee, 2014). Operating at near-zero marginal cost, these network effects are self-reinforcing: early advantage compounds, markets tip irreversibly toward single providers, and the resulting concentration carries welfare consequences incomparably broader than those of a physical monopoly. When a single platform intermediates labour, payments, communications, and commerce simultaneously, its decisions carry a social weight closer to that of public institutions than private firms (Plantin et al., 2018).

Governing this infrastructure as a public good requires demand-side frameworks capturing value generated in downstream uses – value that is real but difficult to price, making open commons management socially desirable even where commercially unattractive (Frischmann, 2005; 2012). This undervaluation is constitutively international: governance arrangements that exclude most countries from standard-setting fail to deliver precisely the most valuable dimension of the public good (Coyle, 2020; 2021; 2025). Data sits at the apex of this value logic – non-rivalrous, compounding in value with

reuse and combination, yet made artificially excludable through intellectual property regimes, access controls, and platform architectures.

The policy question is not whether digital infrastructure has public-good properties but who captures the surplus those properties generate, and under what governance conditions.

The Global Governance Gap

That question has become harder to answer as digitalisation deepens into datafication and both become inseparable from globalisation. National regulatory frameworks cannot govern platforms whose network effects and data flows are inherently transnational – a constitutive feature of the global political economy that incumbent platforms actively preserve (Kaul et al., 1999; 2003). Three governance deficits are acute: the jurisdictional gap between digital infrastructure’s global reach and national regulatory authority; the participation gap reflecting the exclusion of most countries from governance processes infiltrated by Big Tech interests; and the incentive gap between private investment logic and social welfare. The enclosure of data infrastructure by a small number of vertically integrated corporations – simultaneously provisioning infrastructure, operating platforms, and extracting data to develop the most powerful unregulated general-purpose technology in history – is the latest and most consequential form of this enclosure, operating at global scale and at a speed existing multilateral frameworks were not designed to match.

Structural Premises

This paper contends that sustainable digital public infrastructure cannot be realised without confronting five structural realities:

- That the fracturing of multilateral architecture is not a temporary failure but a structural condition, faced from compounded disadvantage by the Global South populations who were never its primary beneficiaries.
- That DPI is deployed in a world of deep, historically produced inequality where each successive technological layer – connectivity, platforms, data, AI – required capabilities that prior layers failed to universalise, making digital inequality self-reinforcing.
- That digitalisation and datafication are not merely sectors requiring governance but are implicated in the conditions that made global governance necessary, with the emancipatory potential of the internet progressively foreclosed by private enclosure that produced a concentration of wealth and influence now actively resisting the governance it requires.
- That the most defensible public good claim within DPI rests not on open-source software and standards but on data itself – non-rivalrous, non-excludable only through active regulatory maintenance, and the critical productive input whose enclosure by platform capital represents the primary governance failure of the digital era.
- That despite its ambiguous promotion by commercially driven technological philanthropy, DPI represents a genuine institutional response to private oligopoly whose potential must be reclaimed through democratic state formation, social compacting, and equitable inclusion.

Cutting across all five is the question of how global collective action can create conditions for the realisation of digital public goods at regional and national levels.

Structure of the Paper

Having set the geopolitical context in Section 1 and the analytical framework in Section 2, above, Section 3 diagnoses the failures of global digital governance since WSIS, culminating in the Global Digital Compact, and argues that data has emerged as the central digital public good whose private enclosure defines the inequity of the contemporary digital economy. Section 4 examines DPI as an institutional response to digital enclosure, interrogating the limits of techno-legal and stack-based governance approaches. Section 5 advances the case for DPI as a trust infrastructure and a project of democratic state formation, legitimacy, and sovereignty. Section 6 examines the financial and regional governance conditions under which transformative DPI could be realised in practice.

3. The Global Governance Record: From WSIS to the Global Digital Compact

The failure of global governance to realise digital public goods is not an aberration but the predictable outcome of a system designed to serve the interests of dominant states and global capital. The liberalisation of telecommunications following GATS in the 1980s disrupted the natural-monopoly logic of national public utilities, producing interconnected global networks that enabled the uptake of the internet – non-rivalrous, federated by design, with no single controlling entity – which appeared as a genuinely democratising global commons.

The first World Summit on the Information Society (2003–2005) foregrounded these public-good characteristics, declaring universal affordable access essential to human rights and innovating new forms of multistakeholder governance (Ostrom, 1990). Yet, over two decades, declarations have failed to yield binding solidarity funding or redistributive financing. The dominant epistemic communities of the World Bank, ITU, and OECD channelled substantial resources through the GATS blueprint – building infrastructure, lowering prices, supplementing universal service funds – yet the digital divide persisted and deepened.

What deepened in response to the realities of the state, no longer the resources or control over public goods, both through privatisation and subsequently private enclosure of data as key input in the economy, was the shift in governance from exclusively state to multistakeholder that produced new conceptions of ‘public’. Best and Gheciu (2014) to explore “disruption and transition” in shifting conceptions of what notions like ‘public’ and ‘private’ mean. “Their conceptualisation of ‘public’ as a set of practices rather than a single institution contributes to the growing literature on the blurring of boundaries between public and private in governance discourse (Cammaerts and Mansell, 2020)” (in Gillwald 2020:41). Together with Hirschman’s (1958) notion of private provisioning of social and public goods that does not exclude an enabling role for the state, these non-state forms of ‘public’ in global governance and in local delivery, provide a path out from the polarised views of the state as public and the market as private that offer insights for digital public infrastructures.

The UN Secretary-General’s High-Level Panel on Digital Cooperation led to the Roadmap for Digital Cooperation (2020), which explicitly centred digital public goods – calling for universal affordable connectivity by 2030, promotion of open-source digital public goods, digital inclusion for the most vulnerable, strengthened capacity building, and the protection of human rights online and offline. The Roadmap reflected the now-rigidly applied equation of digital public goods with ‘openness’ – drawing on Silicon Valley and libertarian tech notions reflected in the UN-associated Digital Public Goods Alliance (DPGA). Together, WSIS and the Cooperation Roadmap represented an explicit global consensus that the internet carries the character of a public good requiring active governance – yet neither generated the binding commitments necessary to realise that aspiration.

Reflecting the extent of deliberation associated with global governance and the lack of measurable outcomes, the Broadband Commission, completing its fifteen-year mission in 2025, recorded over 100 knowledge products, 35 multistakeholder working groups, and 70 policy recommendations – a legacy more legible as process than as outcome. In the same year, drawing on the ‘meaningful connectivity’ surveys conducted in Brazil and South Africa for the G20, research suggested the ITU’s estimate of 2.6 billion people remaining unconnected was closer to double that figure, with several African countries still below the 20–30% penetration threshold needed to realise meaningful network effects (Gillwald, 2024).

Datafication and the Absence of Global Governance

What followed digitalisation was qualitatively different. Datafication – the transformation of social, economic, and political life into extractable, monetisable data – drove platformisation. This transformed the governance challenge entirely. Platforms expanded from intermediary services into foundational infrastructure. Through the commercial enclosure of user-generated data and vertical integration of cloud infrastructure, public values – access, privacy, democratic discourse, public knowledge – were systematically subordinated to the logic of attention capture and behavioural monetisation. The result was unprecedented concentration of power in largely unregulated corporations: GAFAM in the United States and BATX in China. Advanced AI has concentrated power further still, with over 75% of global AI market capitalisation held within two competing blocs contesting control of compute, semiconductors, and rare earth minerals.

Over the past three decades of global governance of digital public goods, digital inequality proved self-reinforcing. At each successive layer – connectivity, platforms, data, AI – the same pattern repeated: each new layer required capabilities that prior layers had failed to universalise (Gillwald, 2017; 2024).

The COVID-19 pandemic made these catastrophic inequalities visible. Billions of people in the Global South were unable to substitute digital access for physical access to work, schools, banking, and social relief – because privatised digital service delivery had failed to build inclusive infrastructure. Simultaneously, the pandemic accelerated corporate capture: emergency conditions enabled technology corporations to colonise public functions across health, welfare, and education, consolidating dependencies that persisted after the emergency passed – ‘sector transgressions’ that deepened state dependency on private actors and raised fundamental concerns for democratic legitimacy (Taylor et al., 2026).

This failure generated the political impetus for the DPI agenda advanced under India’s 2023 G20 presidency, positioning open, interoperable, state-mandated digital systems as a vehicle of post-pandemic reconstruction and explicitly recognising the limits of market provision (examined in Section 4).

The Global Digital Compact and Its Limits

The Global Digital Compact (2024), adopted alongside the Pact for the Future, represented the most explicit integration to date of digital public goods, DPI, and international human rights law – with references to sovereignty and trust that went beyond previous techno-solutionist framings. Yet references to democracy and legitimacy remained conspicuously absent, funding fell short of ambitions, and AI capacity building was resourced at the expense of underlying digital inequality, including digital public statistics, a foundational public good, which were systematically underprioritised (G20 GDC, 2024). While the parallel G20 process under Brazil’s presidency was bolstered by the GDC, and the Marcio Declaration committed to DPI and data governance, the South African G20 presidency was constrained by the chilling effect of the incoming US troika, failing to reach consensus on either transformative DPI or equitable AI governance. The WSIS+20 Resolution (UNGA A/RES/80/173, 2025) recognises DPI as a driver of inclusive digital transformation but remains a normative horizon without operational architecture.

The treatment of digital public goods in both instruments reinforces the Digital Public Goods Alliance (DPGA) definition. The DPGA defines digital public goods as open-source software, open data, open AI models, open standards, and open content, and assesses them against a nine-indicator standard. This functions as threshold certification rather than governance mandate: identifying a category of open products without requiring that resources with genuine public-good characteristics be governed as such. In encoding the equation of openness with publicness into multilateral instruments, it foreclosed precisely the institutional contest over the degree of publicness that effective governance requires (Ostrom, 1990; Kaul et al., 1999).

Open technical standards and interoperability frameworks are more accurately understood as regulatory mechanisms – instruments to prevent enclosure – than as public goods in themselves. They are derivative: governance tools designed to protect underlying resources. Treating them as definitionally equivalent to digital public goods mistakes the instrument for the object of governance and obscures the resource whose enclosure has done the most damage: data.

Data as the Central Public Good

Drawing on the practice of public good theory in policy and regulation, this paper contends that data is inherently non-rivalrous: one person's use does not diminish its availability. Its value compounds with reuse, aggregation, and recombination in its institutional application. The significance of this, however, lies not only in recognising data as the core public good of this era and providing a rationale for its governance, but also in understanding it as the key input into the contemporary economy and society, and, as such, the source of power, wealth, and influence. The value of behavioural and social data derives from the collective contexts that generate it, making data a social product whose enclosure represents not merely market failure but appropriation of social value for private benefit. Without enforceable global governance or, at the very least, guidance reflecting their own conditions and interests, states lacking effective data frameworks are institutionally weakened – unable to generate knowledge for policy, respond to crises, or exercise regulatory authority over actors whose power derives from data exclusivity. Restoring data's public-good character requires not only interoperability mandates, data portability rights, public licensing regimes, and trust frameworks embedded in democratic accountability, but also economic regulation that will counter unprecedented concentration within global markets and redress the current uneven distribution of economic and developmental opportunities associated with advanced data-driven technologies (Gillwald, 2023; GPAI, 2022; Frischmann, 2012; Kaul et al., 1999).

4. DPI: Institutional Response to Global Governance and Market Failure

DPI has emerged as the dominant framework through which states pursue digital transformation. Multilateral institutions, philanthropic foundations, and technology vendors have together constructed a powerful narrative in which DPI adoption is not merely desirable but necessary for development (Burke, 2026).

It gained its principal international momentum through India's 2023 G20 presidency, which positioned open, interoperable, state-mandated digital systems as a vehicle of post-pandemic reconstruction and as a recognition of market failure. The G20 definition that emerged frames DPI as shared digital systems built on open standards, governed by enabling rules to drive inclusion, innovation, and competition while respecting human rights – addressing three foundational layers: digital identification, digital payments, and data exchange, through governance embedded in design and accommodating both public and private participation. India's own trajectory – from Aadhaar through the Unified Payments Interface and the Data Empowerment and Protection Architecture – provided the reference model, along with open-source platforms such as MOSIP, now exported to multiple, mainly African countries.

This agenda did not remain confined to a single presidency. Subsequent G2Os under the Brazilian and South African chairs sustained and deepened the DPI framework, consolidating it as a central feature of post-pandemic economic reconstruction and digital transformation in the Global South. The Global Digital Compact reinforced this trajectory, extending the Cooperation Roadmap's digital public goods framing toward integrated infrastructure stacks capable of delivering public services at scale. What had begun as a nationally inflected model became, within a few years, a multilateral consensus – institutionalised across G20 communiqués, GDC commitments, and a growing ecosystem of prepackaged delivery vehicles including DAAS, GovStack, and the 50-in-5 campaign.

It is at this point that the tensions within the DPI agenda become most visible. A coalition of multilateral institutions, philanthropic foundations, and technology vendors has consolidated a techno-legal framing in which DPI is rendered as a replicable architecture of open standards and enabling regulation, abstracted from the political economies and institutional conditions that determine whether it can function as genuinely public infrastructure. A techno-social approach has emerged in response, insisting that technical architecture is inseparable from governance, institutional design, public value, and inclusion – but this framing, while necessary, remains insufficient without a corresponding account of political economy and democratic accountability (ITforChange, 2025; Aapti Institute, 2024).

Mazzucato, Eaves, and Vasconcellos (2024) sharpen what is at stake: what makes DPI genuinely public is embedding normative principles – accessibility, accountability, non-discrimination, sovereignty – into technical architecture from the outset, not retrofitting them after private lock-in has occurred. Yet the dominant mode of diffusion consistently inverts this sequence. DAAS, for example, acknowledges that implementation realities vary not just between countries but by region, yet proceeds to offer standardised funding for a 180-day DPI rollout framed around common frictions: the tension between need and speed, limited local capacity, funding gaps, and what it characterises as tedious procurement cycles. The heterogeneity of context is named, then set aside.

This is the outcome of depoliticisation. Complex questions of data sovereignty, democratic accountability, and the distribution of value generated through digital systems are recoded as technical problems requiring technical solutions – neutral, scalable, and pre-resolved. The normative ambition of DPI, its promise to embed public values into shared infrastructure, is progressively displaced by an implementation paradigm that prioritises speed of rollout over depth of governance, and treats existing digital ecosystems as integration problems rather than starting points. That displacement is not incidental to how DPI has travelled internationally; it is constitutive of it.

DPI is best understood not as a technical architecture but as an institutional response to compounding governance and market failures – the enclosure of digital infrastructure by private platforms and the massive under provision of foundational digital systems – whose roots lie in structural inequality, and whose consequences the pandemic brought into sharp relief.

The Neutrality Illusion

The dominant framing of DPI as a neutral technical stack is misleading. Design choices embedded in each layer encode political assumptions about identity, consent, institutional authority, and the relationship between individuals, states, and markets – assumptions that are political before they are technical. The corollary is that interoperability, open standards, and modularity are necessary but never sufficient: interoperability enables data flows without determining who controls them; open standards reduce costs without preventing implementation dependencies; modularity allows swapping components without ensuring public-interest governance (Burke, 2025). The two further critiques developed below – on the political economy of the dominant DPI coalition, and on the treatment of fragmentation as a technical rather than institutional problem – follow from this initial recognition.

The techno-legal approach – underpinning India Stack, GovStack for example – treats DPI as requiring simultaneous regulatory and technical design, where code and law are co-constitutive (Massally, Matthan, and Chaudhuri, 2023), but this assumes relatively coherent state capacity as its baseline. In much of Africa, where DPI rollout is primarily targeted, regulatory, technical, and enforcement functions are frequently underfunded, understaffed, or politically subordinate to executive interests. The AU Data Policy Framework, which could anchor DPI governance on the continent, is consistently absent from the blueprints being deployed there.

Individual Safeguards Are Floors, Not Foundations

Privacy and security-enhancing technical standards establish the minimum conditions without which DPI cannot function safely – but they are floors, not foundations. The DPG Standard’s nine indicators represent a minimum normative threshold (Barandiaran et al., 2024). They cannot generate institutional trust at scale, nor substitute for the political and economic conditions that make DPI genuinely public rather than merely technically open.

Manufactured Consensus and Embedded Interests

The breadth of the DPI coalition has generated significant momentum, including through the 50-in-5 campaign, which aims to target 50 countries by 2028. But coalition breadth has required the concept to remain vague enough to accommodate competing visions: from genuinely public digital commons to market-liberalising infrastructure extending platform capitalism into excluded markets. The dominant model – built on India’s Aadhaar and UPI template and diffused through software-industry interests, US philanthropic capital, multilateral institutions, and G20 diplomatic influence – reflects embedded interests that can only be understood within the global political economy. Policymakers, particularly across Africa, are presented with a donor-coordinated model framed as technical best practice rather than a political choice carrying documented exclusion risks. Countries lacking institutional endowments to receive these technologies on their own terms risk deepening rather than reducing dependency.

Technical solutionism to complex problems

Techno-solutionist framings treat fragmentation as inefficiency that can be corrected by better design (Eaves et al., 2024; 2025). The costs are real: the UK reported 44 identity verification systems in 2021; Nigeria’s overlapping ID systems are estimated to have cost \$4.3 billion. But fragmentation reflects the political economy of institutions, in which each system represents patronage, procurement, and bureaucratic power (Acemoglu and Robinson, 2013). Klaaren’s (2023) analysis of

African digital identity systems captures the dynamics that efficiency frameworks mostly obscure. While this might meet some necessary conditions for DPI, they are not sufficient in the African context for several reasons. He argues that assessing the DPI rubric in Africa requires combining the sociotechnical perspective — which recognises that digital identity systems operate within broader social, organisational, and technical relations — with a political economy perspective attentive to industrial structure, competition dynamics, and state capacity. (Klaaren, 2023, citing Whitley and Schoemaker, 2022).

Mazzucato's (2018) critique cuts deeper: where the equation of price with value goes unchallenged, extractive practices present themselves as value creation, manifesting across the Global South as data colonialism — communities whose data powers the system remain excluded from its benefits.

Data as the Core Public Good

The original framing from India's DPI architects treats Digital Public Goods as the bricks and DPI as the building (Kevlani, Ekstep, 2025). This framing directs governance attention toward licensing, standards, and technical openness — and systematically misidentifies where value is created. When openness is treated as the core public good, governance focuses on the wrong layer. When data and data exchange are recognised as foundational, governance must address the structural conditions under which data is collected, held, shared, and used.

Data — administrative, health, financial, environmental, civic — is the foundational public good. Non-rival by nature, its non-excludability is a governance outcome that must be actively maintained. Its social value is realised through combination and analysis across many users and contexts simultaneously. The social return on open public data is enormous yet almost entirely invisible to those who control it, producing the systematic underinvestment Coyle (2020; 2025) identifies as the defining market failure of the digital economy.

Data exchange — moving, combining, and analysing data across systems and jurisdictions — is the infrastructure through which data's public good character is realised. In Frischmann's (2012) terms, it passes the infrastructure test: a shared resource managed for open access, generating downstream social value largely non-commercial and invisible to market mechanisms. Infrastructure that pre-selects which flows it carries directly destroys the public good character of the underlying data.

Open standards are governance instruments, not public goods in themselves — rules maintaining data's public good character by preventing capture through technical incompatibility. Their value is entirely derivative.

DPI systems — identity platforms, payment rails, health information systems — are managed infrastructure whose public good character is conditional: they serve the public interest to the extent they maintain open, non-discriminatory data exchange; they become private platforms to the extent they enclose data or restrict downstream use.

Democratic Decoupling and the Risks of Extractive Deployment

Underlying the global governance failures is the steady democratic decoupling of digital systems: the disconnection of digital infrastructure development, deployment, and governance from mechanisms of democratic accountability. This decoupling operates at multiple levels simultaneously. At the corporate governance level, the decisions that most significantly shape digital environments — algorithmic ranking, content moderation, data monetisation, infrastructure investment — are made by private actors with minimal accountability to democratic processes. At the regulatory level, the technical complexity of digital systems, combined with the systematic defunding of regulatory institutions and the capture of expertise by private actors, has produced a situation in which states often lack the capacity to make informed governance decisions about the systems they nominally regulate (Mazzucato, Entsminger & Kattel, 2021).

At the information environment level, platform architecture has produced epistemic conditions that undermine the conditions for democratic deliberation. The fragmentation of public information environments, the amplification of outrage and division, and the systematic degradation of shared factual foundations are not accidental features of digital communication. They are the product of algorithmic systems optimised for engagement and attention capture rather than democratic discourse (Van Dijck, 2018; Zuboff, 2019).

So, despite the focus on technical failure, some of the greatest risks are through political capture. DPI makes citizens visible to the state – but under extractive institutions, visibility becomes surveillance and state legibility reproduces old domination in a new technical form (Acemoglu and Robinson, 2013). India's experience – the longest available track record – illustrates that even well-designed infrastructure generates contradictions: marginalisation of the most vulnerable, displacement of smaller actors by privileged early entrants, and illegal surveillance (Ashok and Veerebabu, 2023). The standard governance prescription – independent regulators, data governance agencies, interoperability standards bodies – requires institutional endowments built over decades: rule of law, judicial independence, bureaucratic professionalism, civil society capacity. As Klaaren (2023) warns, scaling up DPI without simultaneously building regulatory capacity reproduces the capture dynamics that undermined earlier waves of digital investment.

Moving beyond cost-benefit analysis toward a public value assessment, Eaves et al. propose a three-tier framework for assessing DPI's impact – direct, dynamic, and market-shaping effects. These categories should be treated as a conceptual lens rather than fixed silos, since some direct effects can expand dynamically as DPI adoption grows. This non-linearity matters for Africa, where the sequencing and pace of adoption will vary dramatically across and within countries.

Crucially, the framework has a significant embedded assumption: it presupposes a state with sufficient capacity to measure these effects. The methodological limitations of the evidence base are most acute in Africa, where statistical systems are weakest and where the policy reports that do exist often reflect the framing priorities of international donors rather than the lived experiences of excluded populations.

5. DPI as the Best Available Response: Trust, Legitimacy, and the Social Compact

The UN Secretary-General's call for a Global Digital Compact as part of a renewed social compact (Our Common Agenda, 2021; Pact for the Future, 2024) recognises the same structural gap in the existing global political economy and the absence of global governance. A genuine social compact for DPI binds development and deployment to democratic accountability, redistributive objectives, and the protection of rights – not as external constraints on technical design but as its constitutive purpose (Mazzucato, Eaves, and Vasconcellos, 2024).

Despite the risks, particularly the risk of repeating past development failures by failing to recognise the centrality of democratically accountable institutions, and precisely because of the governance failures documented above, DPI, under the right conditions, remains the most promising institutional pathway for realising the public goods potential of digital systems. The question is not whether to pursue DPI but on what terms, and with what governance architecture.

Trust as the Invisible Infrastructure

Club de Madrid's identification of democratic accountability and effective provision of global public goods as two sides of the same coin captures exactly what techno-solutionist applications miss. Its concept of double-sided causality of trust is foundational: trust enables effective provision of global public goods, and effective provision builds trust. These are mutually constitutive conditions, not sequential steps.

Trust, properly understood, is an institutional property of systems that are accountable, reliable, and governed in accordance with publicly articulated values – the invisible hard infrastructure without which DPI cannot function as public infrastructure at all (African Union Commission, 2023). The failure of several DPI implementations, including aspects of India's Aadhaar system, reflects precisely the consequences of deploying technically sophisticated infrastructure without adequate trust and accountability conditions (Trikanad, 2026). Where trust does not exist, adoption will be shallow: people may enrol when compelled but will not voluntarily extend their engagement, undermining the data exchange that underpins seamless services and public value creation.

DPI as Political Infrastructure: State Formation and the Social Compact

DPI's most significant and underappreciated dimension is its potential contribution to state formation – and its most underacknowledged risk is what happens when that potential is captured by extractive institutions. A reliable digital identity system can extend formal institutional recognition to populations previously invisible to the state, enabling them to access services, assert rights, and participate in formal economic and political life. DPI enabling more effective tax administration, more accurate welfare targeting, and better health and education delivery directly contributes to state formation – generating the legitimacy that sustains fiscal compliance and social trust. The historical pathways through which high-income countries built these capacities were slow, expensive, and often coercive; DPI offers comparable institutional reach at a fraction of the time and cost (Indian Institute of Management Bangalore and Carnegie Endowment for International Peace, 2024).

But this potential is inseparable from sovereign agency: the state's capacity to make autonomous choices on terms accountable to its citizens rather than to external creditors, vendor contracts, or donor conditionalities. DPI is not a shortcut to state formation – it is an expression of it. Where political and institutional foundations are absent, the technical architecture cannot compensate. Where DPI embeds vendor dependency, extracts data under opaque terms, or subordinates public interest to commercial logic, the state's capacity to govern is progressively hollowed out. Financial sustainability and effective control of third-party providers are conditions of sovereign agency, not merely procurement concerns.

The relational dimension compounds this. In much of Sub-Saharan Africa, for example, social protection, credit allocation, land tenure, and dispute resolution operate through relational networks – chiefs, community elders, rotating savings groups, kinship structures – that DPI architectures designed for atomised individuals do not accommodate. Reducing friction in transactions flowing through social intermediaries can simultaneously reduce corruption, destroy social capital, and atomise communities. This is an argument for DPI designed in genuine dialogue with existing social infrastructure rather than imposed as a replacement for it (Coyle, 2025).

This is linked to issues of legitimacy, which operate at three mutually reinforcing levels: input legitimacy asks whether affected populations have meaningful voice in DPI design; output legitimacy asks whether DPI delivers promised services; and throughput legitimacy asks whether governance processes are transparent, accountable, and fair (Schmidt, 2012). Currently, DPI deployments frequently score poorly on all three.

The Virtuous Cycle: Delivery, Trust, Sovereignty, and Public Value

These dimensions converge in a virtuous cycle, reconnecting DPI to the foundational functions of state formation, taxation, social protection, and welfare delivery, which are simultaneously enabled by well-governed DPI and by the prerequisites for it (Loewe et al., 2021). Effective service delivery builds output legitimacy; legitimacy combined with meaningful participation builds institutional trust; trust enables richer voluntary data exchange; richer exchange improves delivery, planning, and resource allocation, generating public value far beyond efficiency gains; and improved delivery in turn deepens state formation and expands sovereign capacity to resist vendor dependency and philanthropic capture. The vulnerabilities in the cycle include: vendor dependency hollowing out sovereign control, exclusionary design producing shallow adoption, governance opacity forfeiting legitimacy, and the absence of a fiscal foundation for sustained public investment. States that cannot deliver services reliably, protect citizens, or include them meaningfully in governance forfeit the legitimacy on which fiscal compliance depends, becoming dependent on external actors and eroding the sovereignty that genuine public value creation requires.

The shadow governance architecture assembled by private actors – through technical design, contractual terms, and automated rule-making – represents the most serious threat, taking consequential decisions about access, identity, and data flows outside the social compact altogether. The governance of DPI as a global public good is not a technical preference but a contestation of the terms on which the digital compact is written. The question beneath every technical debate about DPI is therefore political: what is a public good in the digital economy, who controls it, and who governs the global digital commons, under what terms, and in whose interests.

6. Financial Sustainability and the Transformative Potential of Regional DPI

The enthusiasm for DPI at the level of principle has not been matched by credible financial architectures for its development and maintenance at scale. Most DPI initiatives in the Global South are financed through bilateral donor funding or MDB lending, with limited domestic revenue generation and, as the GDC demonstrated, AI capacity-building commitments came at the direct expense of funding to redress underlying digital inequality.

Structural Underfunding

Structural underfunding reflects three compounding problems. First, the classical GPG financing challenge: non-excludability means providers cannot capture returns, political cycles incentivise national over international provision, and benefits are diffuse and difficult to attribute (Olson, 1965; Kaul et al., 1999). Second, DPI's specific cost structure – high one-time development costs, near-zero replication costs, but substantial recurring maintenance – produces the predictable failure mode in which funding flows into high-profile launches while maintenance is neglected, converting open-source DPI from public goods into security liabilities. Third, DPI's social value is generated in downstream uses that are invisible to market signals and development metrics, leading GPG investments to lose out to more attributable interventions even when social returns are higher (Coyle, 2025). The measurement gap is directly a financing gap: what cannot be measured cannot be budgeted.

Existing mechanisms compound these problems. ODA is poorly calibrated to DPI's GPG character – project-based disbursements favour bounded national interventions over global interoperability. The MDB loan-not-grant bias asks countries to borrow for infrastructure whose benefits are substantially international – economically irrational for heavily indebted economies. A Financial Intermediary Fund model, drawing on Global Fund and Pandemic Fund precedents, offers pooled, grant-based financing with conditionality linked to openness and interoperability. Blended finance carries mission drift risk: infrastructure that begins as open can be progressively enclosed as platformisation dynamics establish lock-in (Frischmann, 2012). Philanthropic capital is insufficient in scale, unaccountable to democratic constituencies, and subject to drift. Domestic resource mobilisation remains the most sustainable foundation – but governance legitimacy is a precondition for fiscal sustainability, not a separate agenda.

The political economy of underfunding is itself part of the problem. The current arrangement – in which the benefits of digital infrastructure are captured primarily by a handful of corporations in two countries while the costs are borne by the populations whose data and attention generate those benefits – is not merely unjust but institutionally unstable. A fair international tax regime for cloud and digital services would provide essential revenue to support state formation, sovereignty, and legitimacy. Instead, the deployment of cloud services, DPI, and AI on extractive terms represents a massive, largely unaccounted for, outflow of public resources – particularly from the African continent.

The Case for Regional DPI

Regional approaches offer the most credible response to the financial sustainability and sovereignty challenges facing smaller economies – combining economies of scale, pooled technical and regulatory capacity, and governance institutions capable of exercising meaningful authority over global digital actors in ways individual states cannot.

The Pan-European model is the most advanced expression of experience, which illustrates this, effectively constituting a collective digital industrial policy through which member states assert sovereignty through at regional scale: harmonised regulation, shared infrastructure standards, and a functioning digital single market have produced what amounts to a collective digital industrial policy through which member states recover sovereign authority over transboundary platforms (CEPS, 2025).

It demonstrates that regional DPI can serve as the institutional architecture through which digital public goods are realised at scale – not a second-best substitute for global governance but a practical pathway to it. This holds particular relevance for Africa.

Few African countries can build a fully integrated DPI stack independently, but through shared infrastructure, harmonised policy, and reciprocal cross-border data flows, a Pan-African DPI becomes feasible – and the AfCFTA provides the foundational framework. The case is made more urgent by a deteriorating external funding environment in which declining aid, shifting donor priorities, and conditionality-laden financing have consistently compromised policy autonomy and long-term sustainability. Regional integration offers a structural alternative: pooled fiscal resources, shared implementation costs, and collectively negotiated terms with global platforms – coordinated through the African Union and anchored in AfCFTA digital trade protocols – could generate sustainable financing streams that are neither aid-dependent nor extractive. Existing international trade frameworks have systematically favoured incumbent digital platforms; regional frameworks that treat infrastructure, data governance, and platform regulation as dimensions of a shared project offer a pathway to rules that serve regional development interests rather than reproducing global asymmetries.

Beyond financing, shared identity, payments, and data exchange layers – developed and governed by African institutions – would reduce reliance on proprietary foreign platforms, retain data value within the continent, and create the conditions for an endogenous African digital economy. Pan-African DPI is therefore not merely a technical or fiscal proposition; it is a strategic instrument for economic self-determination in a world where sovereign agency is increasingly digital.

7. Conclusion: Toward a Trust Infrastructure for the Digital Age

The analysis in this position paper points to a set of interlocking conclusions about what a path out of the current crisis of global digital governance toward a more just, sustainable, and democratically accountable digital order would require. This paper has sought, by tracking the evolution of global governance from digital public goods and infrastructure, to answer the key question of who identifies and chooses to govern global digital goods and how this influences whether there is a genuine digital commons, under what institutional arrangements, and in whose interests – a question made more urgent, not less, by the current crisis of multilateralism and the unprecedented concentration of private wealth and power that datafication and AI have enabled. The exponential private enclosure of critical digital infrastructure has produced an unprecedented concentration of wealth and political power in the hands of a small number of individuals who have reorganised the economy and society in their interests – and who now actively resist the governance that reclaiming it would require. The current arrangement, in which the benefits of digital infrastructure are captured primarily by a few global corporations while the costs are borne by the populations whose data and attention generate those benefits, is not merely unjust but institutionally unstable.

Rebuilding democratic accountability over digital systems, therefore, requires structural transformation, not regulatory adjustment.

This paper has argued that under the right structural conditions, DPI could address longstanding developmental challenges in state formation and public value creation. DPI is not just a technology for service delivery. At its best, it is infrastructure for the institutional inclusion of people who have been systematically invisible to the state – and therefore unable to claim the rights and services to which they are formally entitled. Deployed as a techno-solution – as is currently the dominant mode – without sovereign agency, financial sustainability, or political legitimacy, it instead reproduces the enclosures and dependencies it was designed to overcome.

The question of financing is central here: it is arguably the greatest and most persistent failure in the realisation of digital public goods, and it is directly connected to the deeper questions of sovereign agency and institutional legitimacy. A fair international tax regime for cloud and digital services would provide essential revenue streams to support state formation, sovereignty, and legitimacy. Instead, the deployment of cloud services, DPI, and AI on extractive terms represents a massive, largely unaccounted-for outflow of public resources – particularly from the African continent.

Given the decades of failure to realise digital public goods – and the underdevelopment this has perpetuated across much of the Global South, now compounded by the weaponisation of aid and sovereign funding cuts – regional DPI is not a second-best substitute for global digital governance. For many countries, it is the most credible available path to sustainable, accountable, and democratically governed digital infrastructure. The Pan-European DPI framework is the most advanced expression of this, effectively constituting a collective industrial policy that asserts the digital sovereignty of member states. The Pan-African digital transformation agenda, embedded in the African Continental Free Trade Area, already provides a harmonised framework for data governance, infrastructure sharing, and broadband provision. Its principal limitation – the absence of binding treaties that would allow for mandatory implementation and enforcement – is a political challenge, not a technical one, and one that regional political institutions have both the mandate and the responsibility to address.

Legitimate global institutions capable of governing transborder corporations and global markets – and of ensuring that the benefits of digital public goods are realised at the national level – will not emerge from the diplomatic frameworks that produced the current impasse. They require institutional designs grounded in the political economy of who benefits, who pays, and who is accountable, and a frank recognition that the trust deficit in multilateral governance was not an accident of recent geopolitics but was built into the system's foundations. Arguably, more urgently now, with the first and most powerful general-purpose technology in history concentrated in a few private hands already demonstrating an existential threat to mankind, is what collective action could be taken by whom,

when the actors operate outside international law, global governance, or even the financial system, unlike proposed nuclear non-proliferation- type treatise.

The question Club de Madrid rightly poses – whether a new multilateralism can be constructed that earns trust rather than assumes it – therefore turns on a prior and more fundamental question: whether democratic actors are willing to contest the structural conditions that make trust impossible under the current order. Those conditions are not immutable. But they will not change through technical refinement or incremental regulatory adjustment alone. They require the kind of deliberate, politically accountable institutional construction that has always been the precondition for any durable public good – and a recognition that the digital public goods the world urgently needs will not be delivered by the market arrangements, diplomatic frameworks, or governance architectures that foreclosed them in the first place.

** Claude Sonnet 4.6 was used in the initial compilation of the author's published and unpublished work. It was also used to condense sections of earlier versions of the report, specifically to identify financing mechanisms beyond those that have historically arisen in the global governance of digital public goods.*

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Annex: Financing Proposals

Global Solidarity Fund

- Collection of primary digital public goods, public statistics, and the CETIC funding model proposed by the South African G20 Presidency
- Digitalisation of existing registries and public sector data
- Seville Development Funding: fill existing underlying broadband and power infrastructure; demand-side support and public value social investment in education and health
- Mobilise domestic capital (and regional-level resources) for shared infrastructure and to support common markets
- Contribution design: base schedule linked to digital economy GDP share for major economies; voluntary top-ups; feasibility exploration of a Digital Economy Solidarity Levy
- MDB DPI windows with grant elements
- Institutional development – multi-year operational support, not capital-only financing, directly addressing the maintenance gap

Digital Economy Solidarity Levy (Medium-Term)

- Commission a feasibility study for a small levy on cross-border digital transactions or digital advertising revenue, hypothecated to the Global DPI Fund. Modelled on the Tobin
- Tax concept; draws on Kaul's (2013) GPG-generated fiscal instruments analysis. Even 0.01–0.1% on relevant transaction volumes would generate substantial revenues.
- Medium-term objective requiring sustained political coalition-building in the G20 digital economy track and international tax reform processes
- Proposed mechanism for digital public statistics

ODA Classification Reform (Near-Term)

- Advocate with OECD DAC for clarification that contributions to global DPI standards bodies and open-source DPI codebases count as ODA – lowering the political cost of bilateral donor contributions to multilateral DPI governance and enabling accurate tracking of official finance flows to DPI as a GPG.

DPI Finance Transparency Mechanism (Near-Term)

- Public DPI Finance Tracker (model: Climate Finance Transparency Initiative) to monitor financing flows globally, identify gaps by component type and recipient region, and hold donors accountable for financing commitments. Integrated with DPI Coordination Mechanism.

